Vehicle Rental SaaS — Vehicle & Document Management Specification

Production specification covering vehicle creation, locations, document management, compliance, inspections, maintenance, pricing, rental rules, and related fleet modules.

1. Add Vehicle — Main Sections

#	Section	Main purpose
1	Basic Information	Identity, registration, category, status
2	Specifications	Technical and physical details
3	Pricing	Rates, deposit, taxes and extra charges
4	Locations	Branch, parking, pickup/return and transfers
5	Photos & Media	Fleet and inspection images
6	Documents	Registration, insurance and compliance files
7	Maintenance	Service schedule, work orders and downtime
8	Rental Rules	Eligibility, mileage, fuel and booking rules
9	Initial Inspection	Condition, odometer, fuel and damage
10	Ownership & Finance	Ownership, purchase and supplier information
11	Review & Publish	Validation before making the vehicle rentable

2. Basic Vehicle Information

- Vehicle display/internal name
- make
- model
- model year
- category
- registration/plate number
- VIN/chassis number
- engine number where applicable
- color
- internal fleet ID
- status
- date added.

3. Vehicle Specifications

- Fuel type (Petrol/Diesel/EV/Hybrid/CNG/LPG where applicable)
- transmission
- engine CC
- EV battery capacity
- seats
- doors
- body type
- odometer

- fuel tank/battery range
- AC/GPS/Bluetooth/ABS/airbags and other features
- bike/scooter helmet count.

4. Pricing & Charges

- Hourly/daily/weekly/monthly prices
- minimum/maximum rental duration
- security deposit
- late fee
- extra-hour fee
- extra-km fee
- taxes
- seasonal/weekend pricing
- one-way fee
- discounts
- optional extras.

5. Locations & Fleet Movement

- Primary branch
- current physical location
- parking/storage location
- pickup location
- return location
- allowed pickup/return branches
- one-way rental
- vehicle transfer history
- transfer date/time
- who moved it and why.

6. Photos & Media

- Main photo
- front/rear/left/right
- interior
- dashboard/odometer
- fuel/battery level
- damage photos
- marketing gallery
- ordering/primary image
- upload metadata. Store large files in S3-compatible object storage rather than PostgreSQL.

7. Document Management — Dedicated Subsystem

- Document type
- document number
- issuing authority

- issue date
- expiry date
- uploaded file
- status
- verification status
- verified by/at
- notes
- reminders
- entity linkage
- version history
- archive/soft-delete
- audit history.

8. Vehicle Documents

- Registration/bluebook equivalent
- insurance
- roadworthiness/fitness
- pollution/emission where applicable
- permits
- road tax evidence
- ownership/lease documents
- import/customs records where applicable
- inspection certificates
- other regulatory documents.

9. Document Management Features

- Central Documents page
- search/filter
- valid/expiring/expired/pending/rejected states
- upload/preview/download/archive
- file validation
- versioning
- verification workflow
- configurable 30/15/7-day reminders
- bulk upload
- permission-controlled access
- audit logs
- dashboard widgets for expiring/expired documents.

10. Maintenance & Service

- Maintenance status
- last service date/odometer
- next service date/odometer

- service interval
- preventive maintenance
- work orders
- workshop/provider
- parts/labour costs
- downtime
- notes
- history
- attached invoices/documents.

11. Inspection & Condition

- Inspection date/time
- odometer
- fuel/battery
- exterior/interior
- tyres/wheels
- lights
- brakes
- engine/mechanical condition
- accessories
- existing damage
- damage photos
- inspector
- customer-visible/internal notes
- checklist.

12. Rental Rules

- Online booking eligibility
- customer verification
- driving licence requirement
- minimum/maximum age
- licence categories
- mileage policy
- included kilometres
- extra-km rate
- fuel policy
- rental duration
- advance booking limit
- cancellation rules
- deposit requirement.

13. Ownership & Finance

- Owned/leased/financed/partner

- purchase date
- purchase price
- supplier/vendor
- lease dates
- internal asset value
- accounting reference
- internal notes. Restrict sensitive fields by permission.

14. Operational Status

- Available
- Reserved
- Rented
- Pickup in progress
- Return inspection
- Maintenance
- Damaged
- Unavailable
- Blocked
- Retired. Every important status change should be validated by the backend and recorded.

15. Vehicle Detail Page

- Overview
- current status
- availability
- current location
- pricing
- photos
- documents
- maintenance
- inspections
- damage
- rental history
- bookings
- expenses
- revenue summary where permitted
- activity timeline
- audit history.

16. Other Related Modules / Places

- Locations/Branches
- Vehicle Categories
- Documents
- Maintenance

- Inspections
- Damage Management
- Fuel & Mileage
- Availability
- Pricing
- Bookings
- Customers
- Contracts
- Expenses
- Notifications
- Audit Logs
- Reports.

17. Recommended Database Entities

- vehicles
- vehicle_categories
- vehicle_locations
- vehicle_photos
- vehicle_documents
- document_types
- document_versions
- document_verifications
- vehicle_pricing
- vehicle_rental_rules
- vehicle_inspections
- damage_records
- maintenance_records
- maintenance_work_orders
- fuel_records
- odometer_readings
- vehicle_transfers
- vehicle_status_history
- vehicle_activity_logs

18. Document Record — Minimum Data

Field	Purpose
id	Unique document ID
tenant_id	Tenant isolation
entity_type / entity_id	Linked vehicle/customer/contract/etc.
document_type_id	Configured document type
document_number	Official number
file_key	S3/object-storage key
file_name / mime_type / file_size	File metadata
issue_date / expiry_date	Validity dates
status	Valid/Expiring/Expired/etc.
verification_status	Pending/Verified/Rejected
verified_by / verified_at	Verification audit
version	Document version
notes	Internal notes
created_at / updated_at	Timestamps
deleted_at	Archive/soft delete if used

19. Security & Multi-Tenant Rules

- Never trust company_id/tenant_id supplied by the client; derive tenant from authenticated context and membership.
- Every tenant-owned vehicle, document, photo, inspection and maintenance query must be tenant-scoped.

- Private document files must be protected by authorization; do not expose permanent public URLs.
- Validate file type, size, entity ownership, tenant and permissions on the backend.
- Sensitive ownership/financial documents require granular permissions.
- Important changes must create audit records.
- Expiry-reminder jobs must be idempotent and retry-safe.
- Frontend validation improves UX; backend validation remains authoritative.

20. Frontend Navigation

- Vehicles → All Vehicles
- Vehicles → Add Vehicle
- Vehicles → Categories
- Vehicles → Documents
- Vehicles → Maintenance
- Vehicles → Locations
- Vehicle Detail → Overview / Documents / Maintenance / Inspections / Damage / Rentals / Activity
- Documents → All / Expiring / Expired / Pending Verification
- Locations → Branches / Pickup Points / Transfers / Business Hours

21. Category-Driven Required Fields

Do not make every field mandatory. Vehicle Category should control conditional fields. EVs can require battery data, combustion vehicles can require engine/fuel data, cars/vans can require doors, and bikes/scooters can require helmet-related fields. The backend must enforce the final rules.

22. Final Add Vehicle Checklist

- ■ Basic identity and registration
- ■ Category-specific specifications
- ■ Pricing, deposit and taxes
- ■ Branch/current location
- ■ Photos and media
- ■ Required documents
- ■ Document expiry and verification
- ■ Maintenance schedule
- ■ Rental rules
- ■ Initial inspection
- ■ Existing damage
- ■ Ownership/financial information
- ■ Review and backend validation
- ■ Publish/activate vehicle